

AI Engineering (Week 1)

aie26

Topic 01 : Module Overview

Part 01 : Module Introduction

Motivating Example

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Preparation

Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Autumn Semester, 2026

Outline

- Delivery and Assessment
- Introduction, definitions and context
- Roles, expertise and ethics
- Workflow and process models
- Overview of Machine Learning Algorithms
- Resources

wrap up

AI Engineering (Week 1)

Introduction



Motivating Example

Preparation

Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Prediction

Regression
1

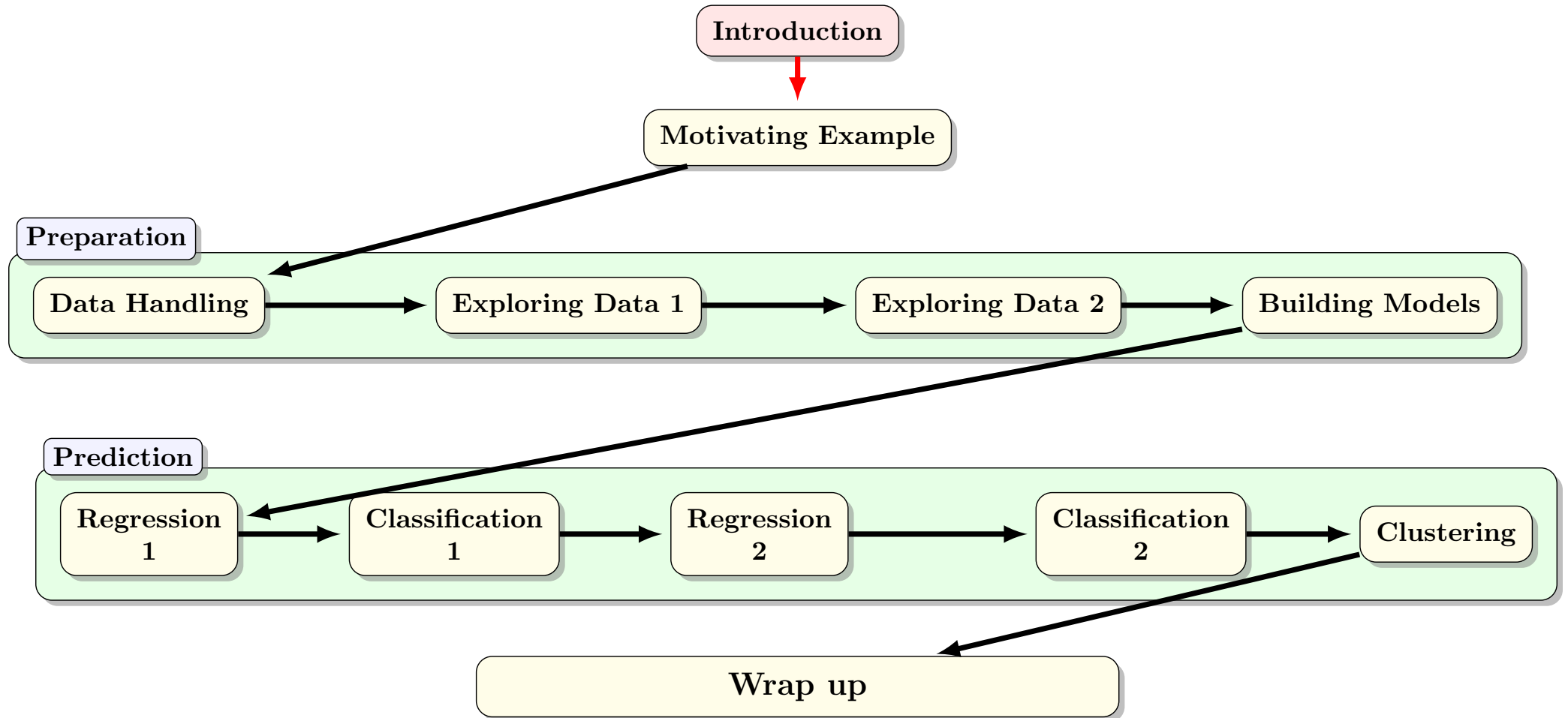
Classification
1

Regression
2

Classification
2

Clustering

Wrap up



Outline

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How? — Delivery

Contact hours - but subject to change!

- 2 hours of lectures per week (all students).

—Tuesday 12:15-13:15 (Room W13)

—Thursday 14:15–15:15 (Room TL252)

- We cover concepts, definitions, examples, etc.
- Lecture objective is to improve understanding of the topic.
- Practical/Lab is used to develop practical experience of an integrated set of topics
- Feel free to ask questions.
- A 2-hour practical/lab session per week for each Group (Room D04).
 - Tuesday 13:15-15:15 - Group W1 (D04)
 - Thursday 15:15-17:15 - Group W2 (D04)
 - Labs use python workbooks to define and implement data mining *workflows*.

(Hardware) Requirements

- Use of own laptop (minimum of quad-core CPU and 8GB RAM is recommended)

How? — Assessment Structure

100% Continuous Assessment

- 80% — two data investigations
 - Issued Week 3, submitted end Week 8 (Data Investigation, 40%)
 - Issued Week 8, submitted end Week 13 (Data Investigation, 40%)
- 20% — Labs and moodle quizzes

These are indicative and might change (in terms of when assignments are issued and their relative weightings)...

What is the AIM of the module?

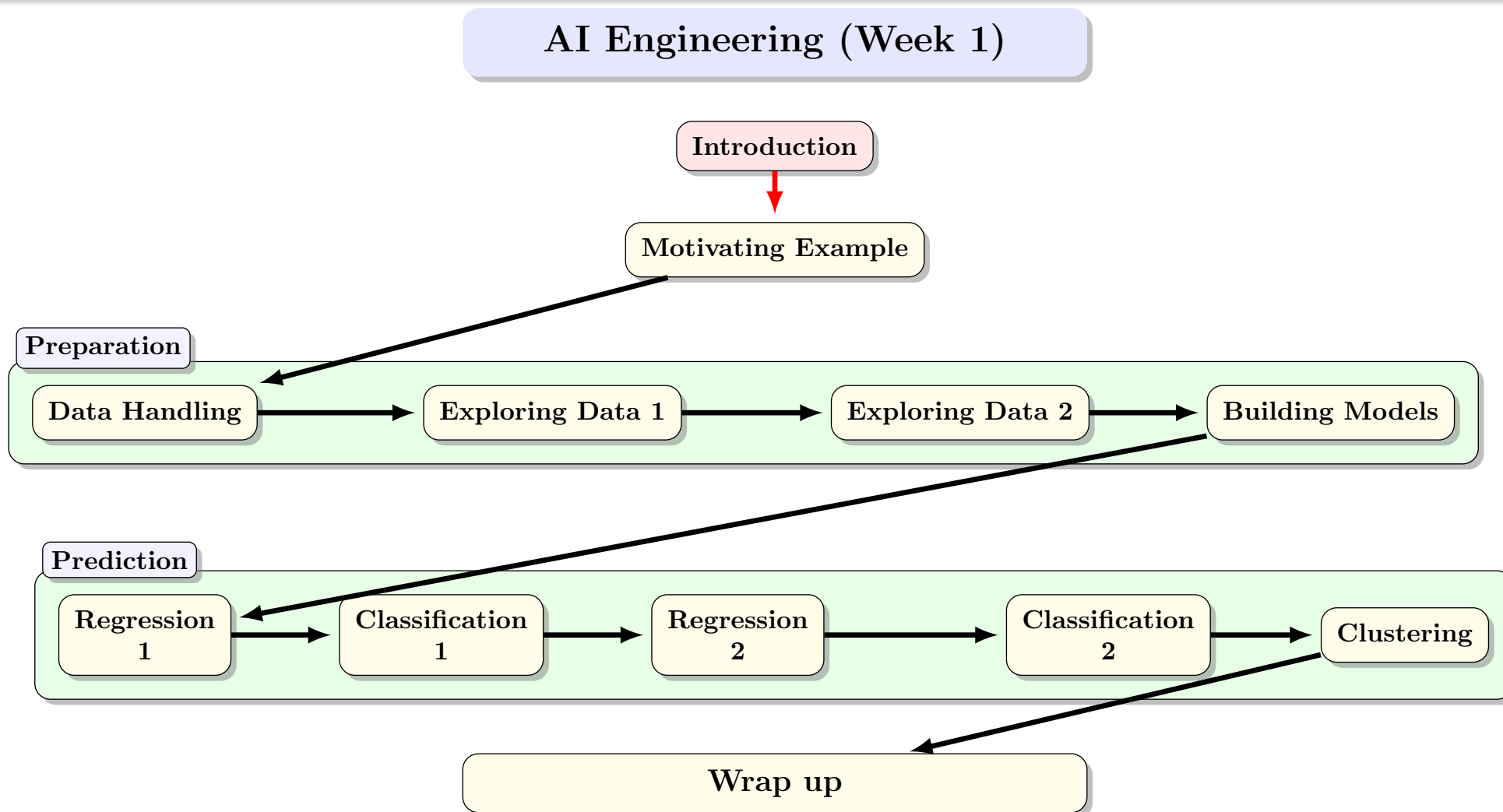
Aim, as per Module Descriptor*...

The purpose of this module is to introduce the student to the fundamental concepts and techniques of Machine Learning. The student will become familiar with both supervised learning (such as regression and classification) and unsupervised learning (such as clustering and dimensionality reduction)...

Translation (Informal Aims)

- ① Collect observations from a variety of processes, yielding large amounts of data.
- ② Preprocess this data, selecting relevant features only.
- ③ Use data-intensive analysis techniques to obtain insights.
- ④ Postprocess analysis results, validate, visualise and refine the process.

What topics does it contain?



Selected definitions

Data Scientist: can ask the right questions, {generate} and consume the results of analysis of Big Data effectively.

(McKinsey 2011)

Machine Learning: Branch of computer science {and related fields} that gives computers the ability to learn without being explicitly programmed. (Samuel 1959)

Deep Learning: Use of very large neural networks with many layers of “neurons” that can be trained to generate robust models of their input, whose classification performance scales with the amount of data supplied. (Various)

Generative AI: describes algorithms that can be used to create (remix?) new content, including audio, code, images, text, simulations, and videos, (based on its training data) (McKinsey 2024)

Artificial Intelligence (AI): the capability of a machine to *imitate* intelligent human behavior (Webster 2017)

Artificial Narrow Intelligence (ANI): also known as weak AI, (exists today) can be trained to perform a single or narrow task, often far faster and better than a human mind can (IBM 2025)

Artificial General Intelligence (AGI): also known as strong AI, is the (currently hypothetical) intelligence of a machine that can accomplish any intellectual task that a human can perform (Gartner 2024)

Artificial Super Intelligence (ASI): If ever realized, Super AI would think, reason, learn, make judgements and possess cognitive abilities that surpass those of human beings (IBM 2025)

More definitions...

Large Language Models: LLMs are a category of deep learning models trained on immense amounts of data, making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks. LLMs are built on a type of neural network architecture called a transformer which excels at handling sequences of words and capturing patterns in text. (IBM, 2026)

Agentic AI: Agentic AI is an autonomous AI system that can act independently to achieve pre-determined goals. Traditional software follows pre-defined rules, and traditional artificial intelligence also requires prompting and step-by-step guidance.

... Agentic AI is proactive and can perform complex tasks without constant human oversight. (AWS, 2026)

Retrieval-Augmented Generation: RAG is a technique for enhancing the accuracy and reliability of generative AI models with information fetched from specific and relevant data sources. ... when users need authoritative, source-grounded answers rather than broad knowledge alone, RAG can provide the necessary depth and accuracy. (NVIDIA, 2026)

Context engineering: Context engineering is replacing prompt engineering as the new frontier of control. It's not about clever wording anymore. It's about designing environments where AI can think with depth, consistency, and purpose. ... we're moving from asking smart questions to building smarter worlds for models to inhabit. (KDnuggets, 2025)

➤ This module focuses on fundamental concepts and understanding, so it just touches upon these topics. ➤

What is the purpose of Machine Learning?

Operational Definitions

- Machine learning builds models to capture underlying patterns in data so it can make accurate inferences on new, unseen data without explicit hard-coded programming.
- models derive knowledge from large and/or complex datasets, with *guidance* from the data scientist
- “Machine learning draws from the disciplines of programming and mathematics/statistics.”

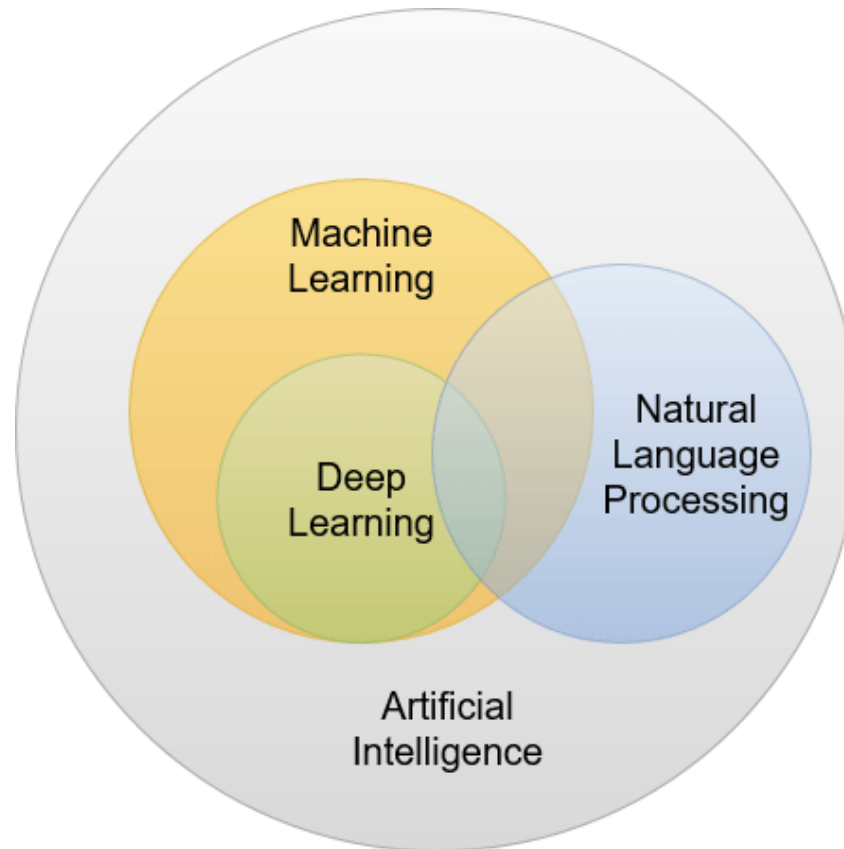
Primary goals

- From messy and noisy raw data, deriving structure and context
- Applying scalable learning algorithms to these higher value data sets

Secondary goals

- Modelling and understanding the error and other consequences of the modelling process.
- Building data-driven processes, architectures & frameworks: *Big Data*

AI vs ML vs DL



- Note the $AI > ML > DL$ hierarchy
- Generative AI (cf., ChatGPT) is very active at the moment, and goes beyond ML and DL.

Interlude: Examples of Big Data

Exercise

please consider (real world) processes generating *Big Data*.
Can you come up with 3 examples in 3 minutes?

Analytics and Classical ML

Data Generation

- Financial transactions (bank, retail, ...)
- Social engagement (texts, comments, likes, shares, ...)
- Machine-to-machine (IoT, cyberphysical systems, ...)
- Content curation and recommendation

Data Processing

- From Relational to NoSQL, to Multi-model databases
- From stored procedures to microservices and serverless computing
- From in-database to Apache Hadoop and Spark
- From proprietary statistical software to R and to python
- From consultants, to data scientists to machine learning engineers

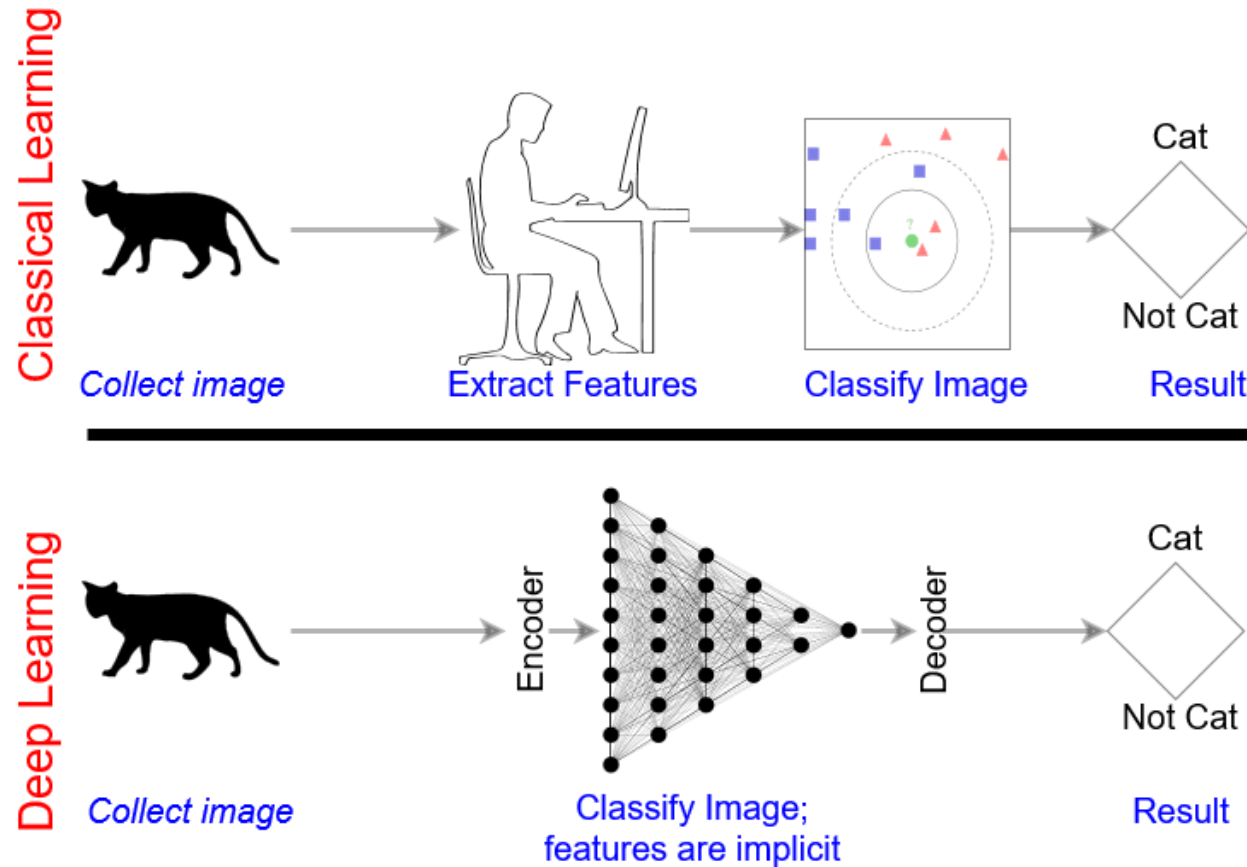
Data Analysis

- Reporting with descriptive statistics and simple plots, to classical ML with predictive analytics
- From predictive accuracy, to robustness vs fragility, to data understanding and visualisation
- Traditional chatbots are limited to specific contexts

The current (generative) wave: 2022–date

- Traditional ML (big data processing for predicting labels or numbers) becomes mainstream
 - ML skills become expected
 - ML is operationalised by incorporating it in other practices (ML + DevOps = MLops)
- *Generative AI* becomes practical
 - Large Language Models for generating textual responses to prompts, e.g., ChatGPT
 - Generate images and other rich media given textual descriptions, e.g., DALL-E
 - Beyond LLMs and next token prediction: RAG, Large Reasoning Models, AI Agents, ...
- AI is beginning to deliver on the promise identified by John McCarthy and Alan Turing and others in the 1950s.

Classical versus Deep Learning - overview



Classical learning requires extensive setup before training.

Classical versus Deep Learning - pros and cons

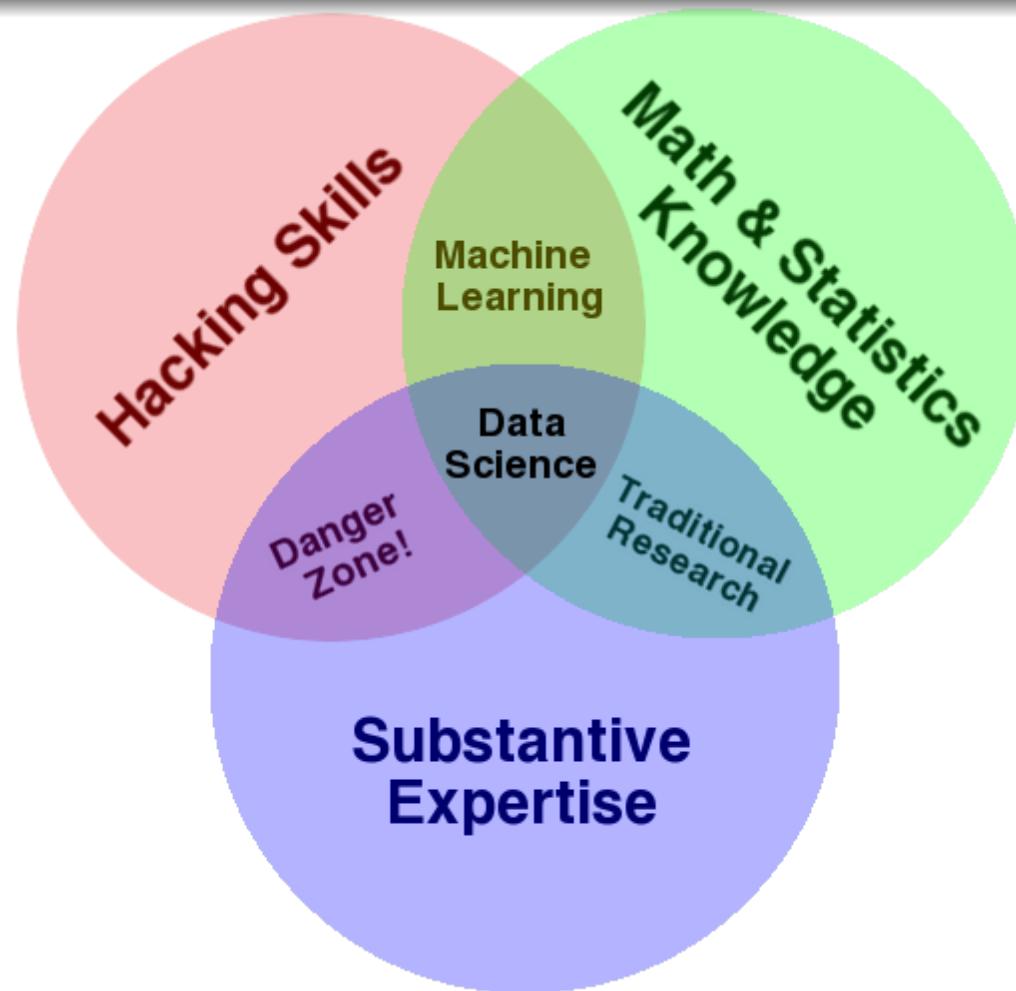
Classical Learning	Deep Learning
Can work well with less data (<10 ⁶ rows, say)	More training data gives more accuracy (within reason!)
Easier to interpret/explain models	Model is opaque and can be fragile
Training is relatively fast	Training can require many epochs
Training requires fewer resources	Training requires massive resources
Accuracy improvement falls off	Accuracy can improve with more training data
Features engineering is performed by humans	Features are encoded implicitly in layers during training
Complex prediction requires complex model	With enough nodes, DL can represent any function

➤ In conditions where one type of learning is weak, the other is often strong.

Emphasis of this module

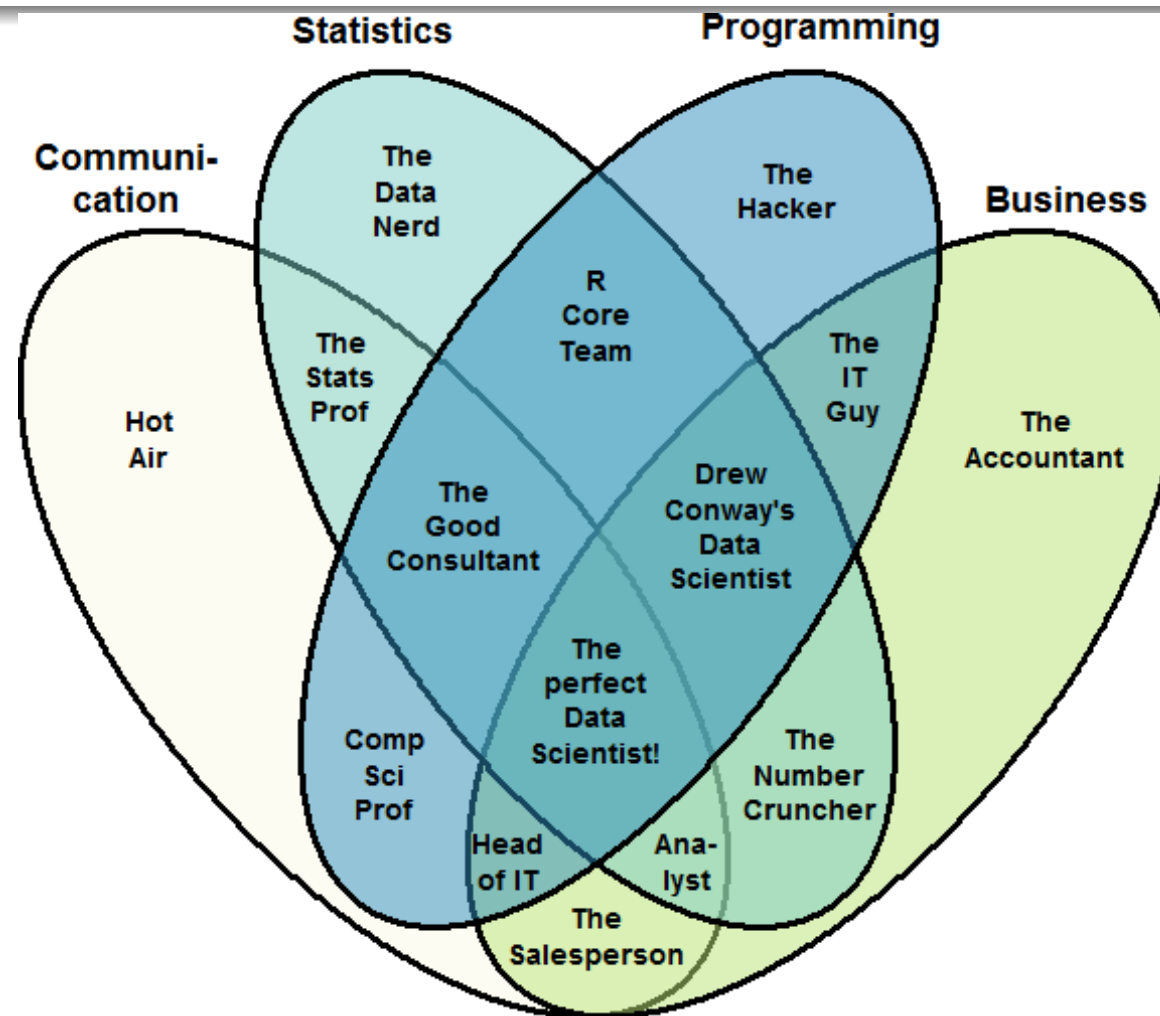
- This module covers foundations, classical models, some deep learning
- Foundations include EDA, notions of error and variance, training vs test data, ...
- Classical modelling includes feature selection and classical approaches to learning from data
- Neural network and Deep learning go straight to predicting based on (encoded) data
- Foundations are shared by both classical and deep learning
- **Deep learning is ideal for learning from labeled, web-scale big data.** Specialised deep-learning models are the basis of LLMs.
- Day-to-day, classical machine learning is often more suitable for predicting from relatively small datasets.

Drew Conway's 3-set Venn Diagram of Data Science Expertise



Source: <http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagram>

Stephan Kolassa's 4-set Venn Diagram of Data Science Expertise



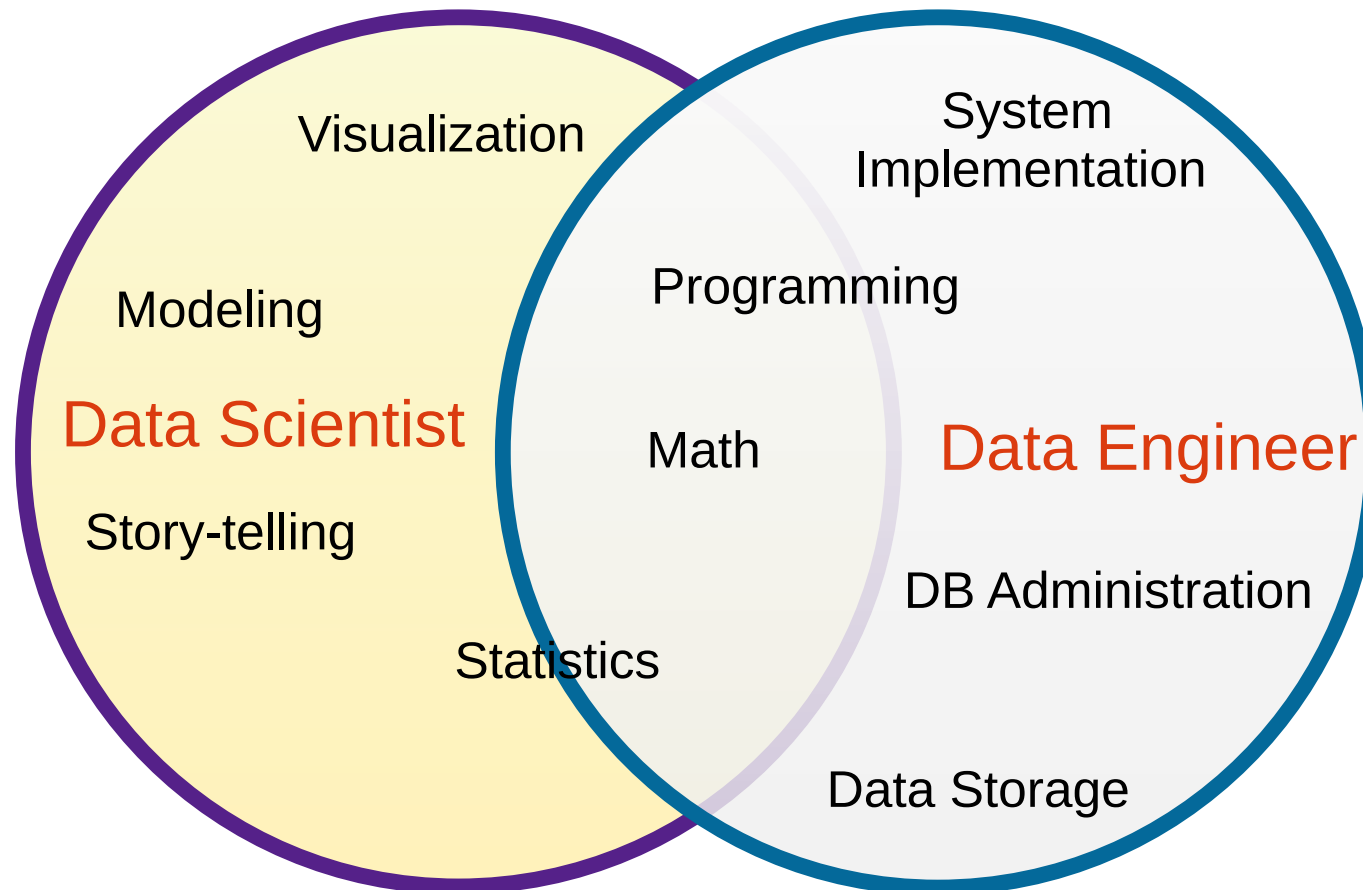
Source: <https://datascience.stackexchange.com/a/2406>

Gartner suggests the need for a *Citizen Data Scientist*



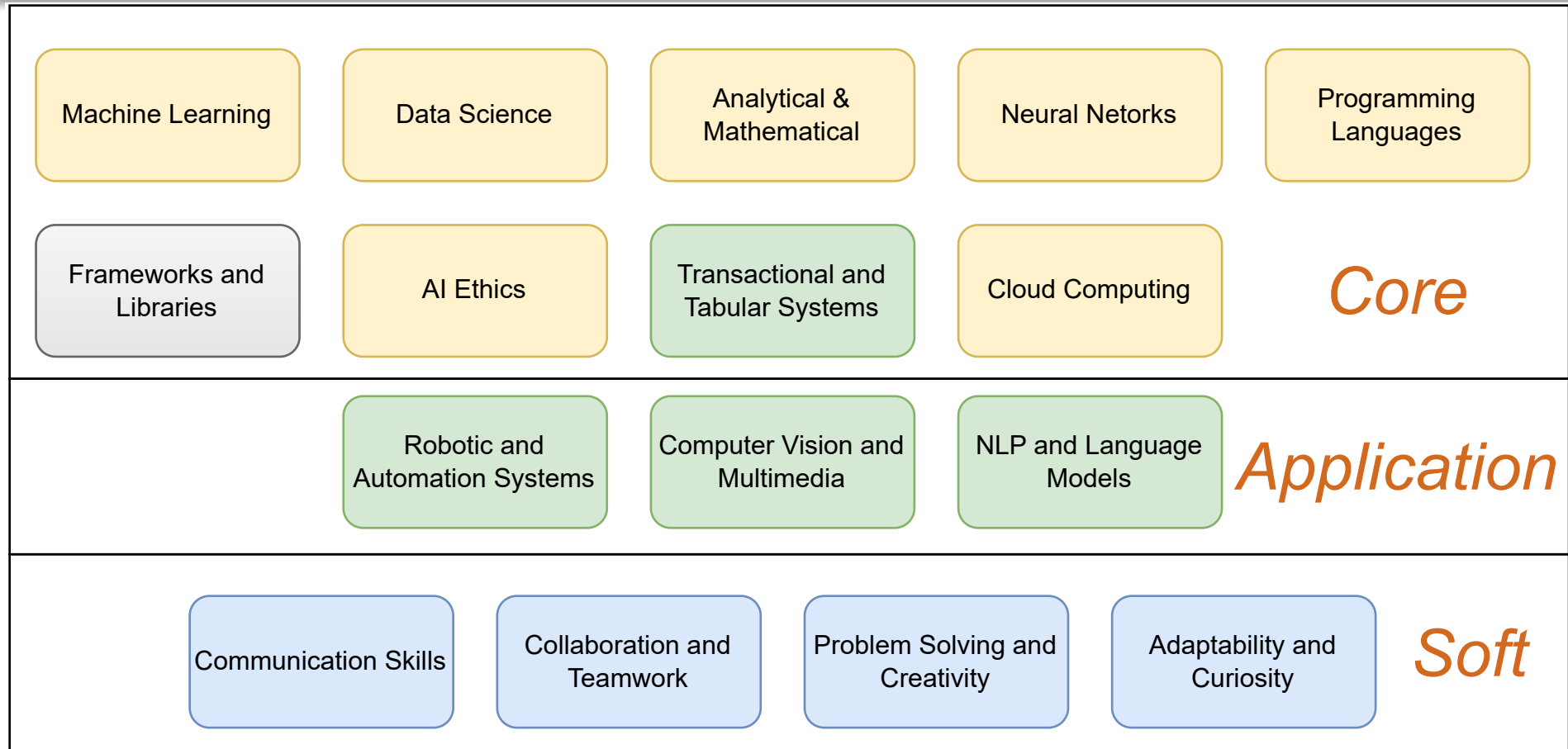
Source: <http://www.kdnuggets.com/2016/03/cartoon-citizen-data-scientist.html>

Data Scientist vs Data Engineer



Source: <https://ryanswanstrom.com/2014/07/08/data-scientist-vs-data-engineer/>
Also the traditional roles of *Data Analyst* and *Software Engineer*...

AI Engineer Skills 2025



Source: Based on <https://dmletterstudio.com/ai-engineer-skills-needed>

➤ This module touches on most of the core skills and some of the soft skills ➤

Ethical Concerns

- protecting privacy
 - obtain informed consent for training data
 - pseudonymisation can be insufficient
- ensuring transparency of decisions
 - opaque models hide how predictions/decisions were made
 - interpretable/explainable models should be used where possible
- breaking cycles of bias
 - Biased training data leads to biased outputs from models
 - Diverse and representative training data gives more reliable predictions
- enabling validation
 - Poor choice of techniques, settings and model parameters can harm prediction quality
 - Careful validation can assess whether predictions are as good as expected
- enabling decisions to be challenged
 - Related to interpretability, predictions without understanding can be misleading
 - Need to support an audit trail, validation procedure and due process

Ethical frameworks and legislation

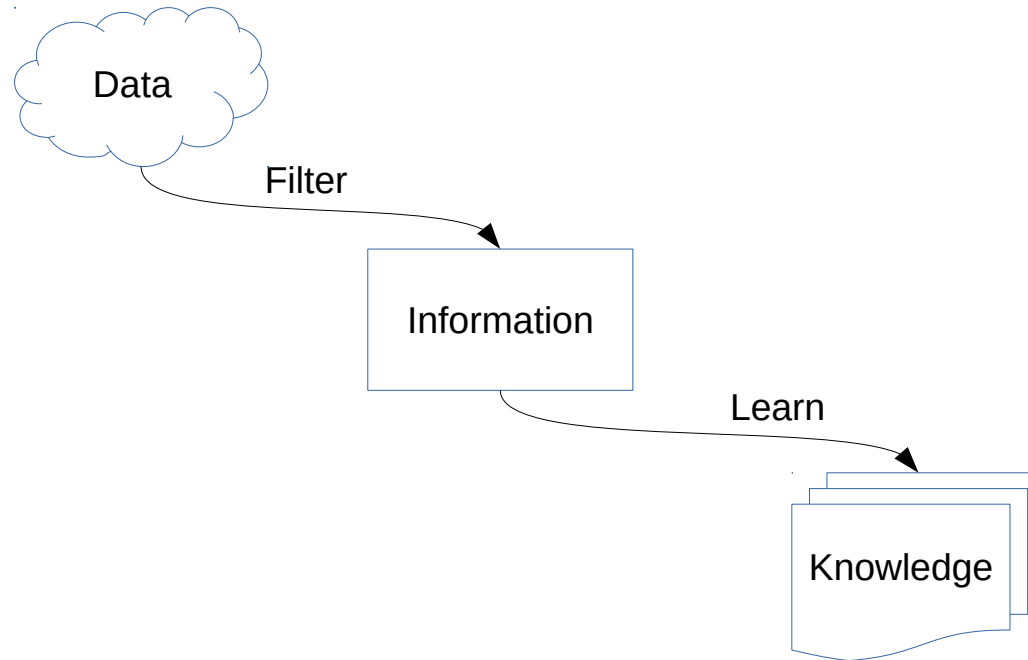
UNESCO human rights approach

- ① Proportionality and Do No Harm
- ② Safety and Security
- ③ Right to Privacy and Data Protection
- ④ Multi-stakeholder and Adaptive Governance & Collaboration
- ⑤ Responsibility and Accountability
- ⑥ Transparency and Explainability
- ⑦ Human Oversight and Determination
- ⑧ Sustainability (c.f., SDGs)
- ⑨ Awareness & Literacy
- ⑩ Fairness and Non-Discrimination

EU AI Act 2025

- Rules for each risk class
- Need for transparency
- Encouraging innovation

The Data to Knowledge Pipeline



Data Filtering

- Clean (drop unwanted observations)
- Summarise (remove observation detail)
- Reduce (remove/transform variables)

Learning

- Derive models
- Validate models
- Analyse discordance

Data - Information - Knowledge - Wisdom

Example of the DIKW chain

- Servers and applications log events in files and/or databases [DATA]
- *Collector* agents select specific events, in context [INFORMATION]
- Machine learning *classifiers* learn system behaviour and identify anomalies [KNOWLEDGE]
- Humans and software use this knowledge to prevent future problems [WISDOM]

Note that the DIKW chain is often represented as a pyramid.

Cross Industry Standard Process (for) Data Mining

CRISP-DM

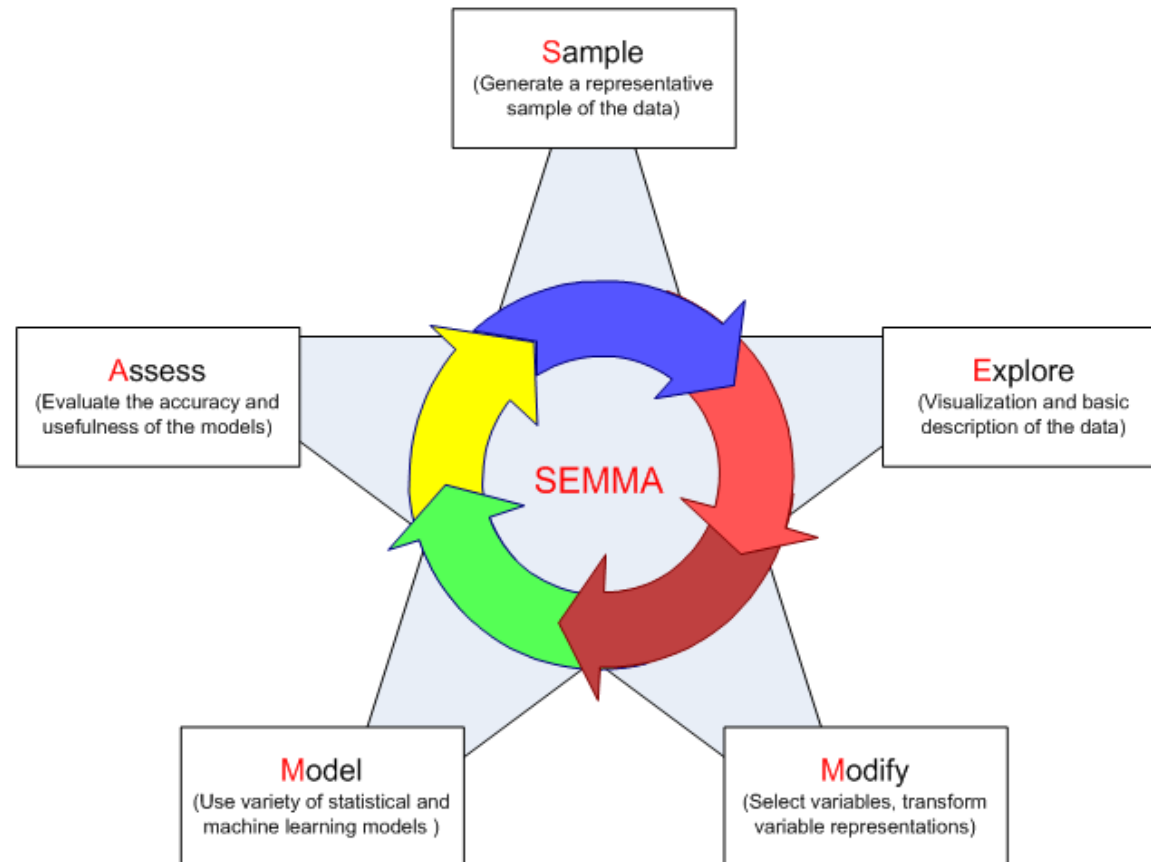
CRISP-DM is a high-level iterative process model. It gives much weight to data understanding and preprocessing and involves the data and problem owners from the start (c.f., Agile). It is the best known but predates Big Data, etc.



Sample, Explore, Model, Modify, Assess

SEMMA

SEMMA is promoted by SAS and takes a more operational view of data mining, using a (statistical) *model-building* metaphor. Business input is essential but largely implicit. It is more concrete than CRISP-DM so it tends to map well to DM tool workflows.

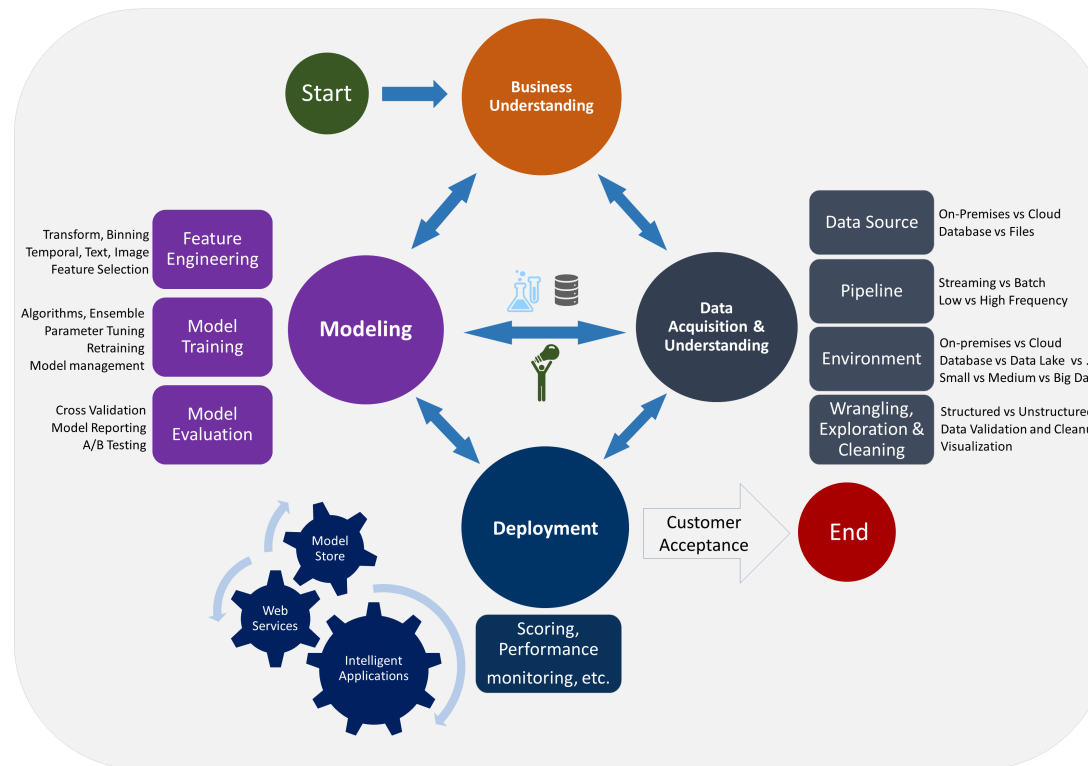


Microsoft Team Data Science Process

TDSP

TDSP is the most detailed process model of the 3. It is much more recent. It is cloud-aware and directly references Azure and other Microsoft technologies. Typically there are two main cycles, one involving the Business, the other involving Deployment. Interestingly, there is a Start and End, so it is more project-focused.

Data Science Lifecycle



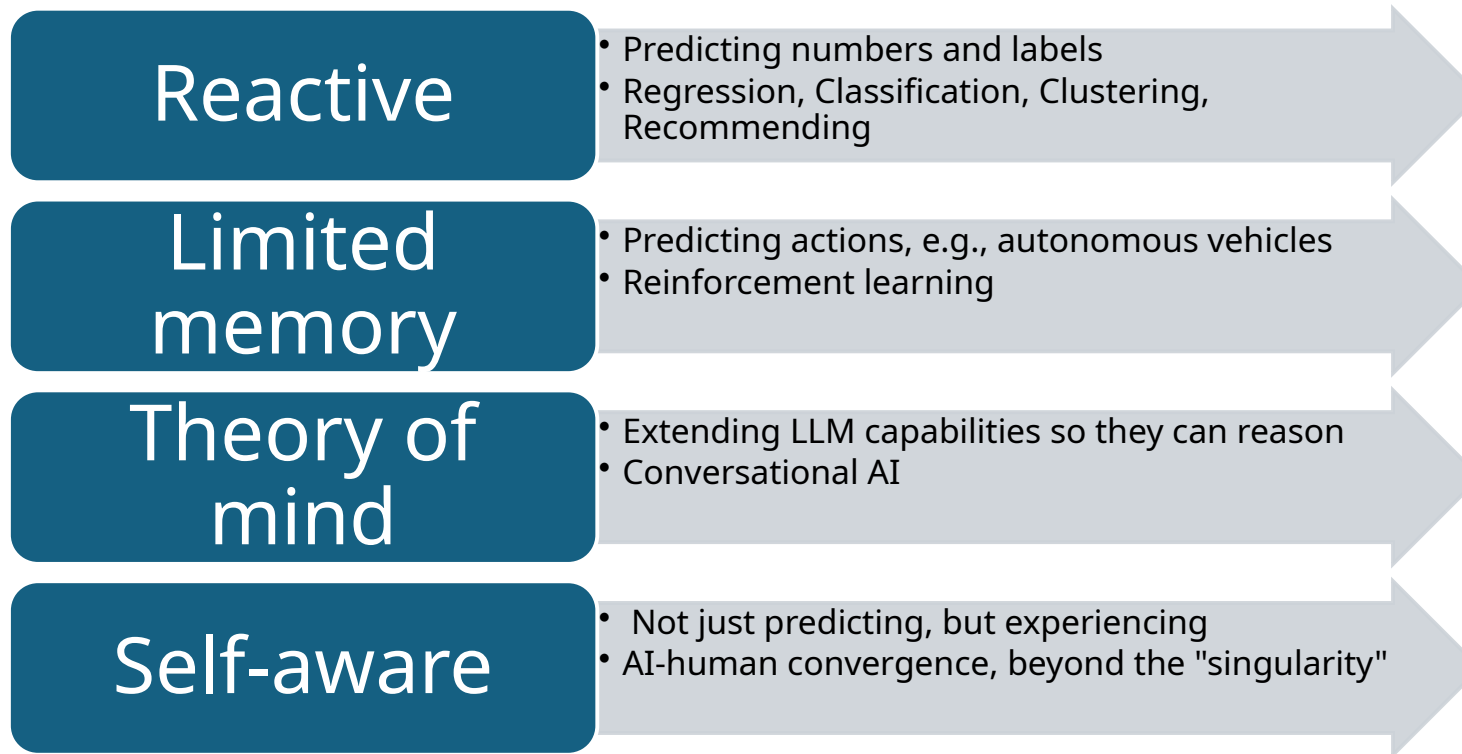
The “5 Tribes of Machine Learning”

Tribe	Origins	Learning Algorithm
Symbolists	Logic, Philosophy	Inverse Deduction
Connectionists	Neuroscience	Backpropagation
Evolutionaries	Mathematical Biology	Genetic Programming
Bayesians	Statistics	Probabilistic Inference
Analogizers	Psychology	Kernel Machines

Summarised from Domingos (2015) “The Master Algorithm”

Each approach works well in specific contexts - there is no “master algorithm” yet...

Bernard Marr's 4 Types of AI

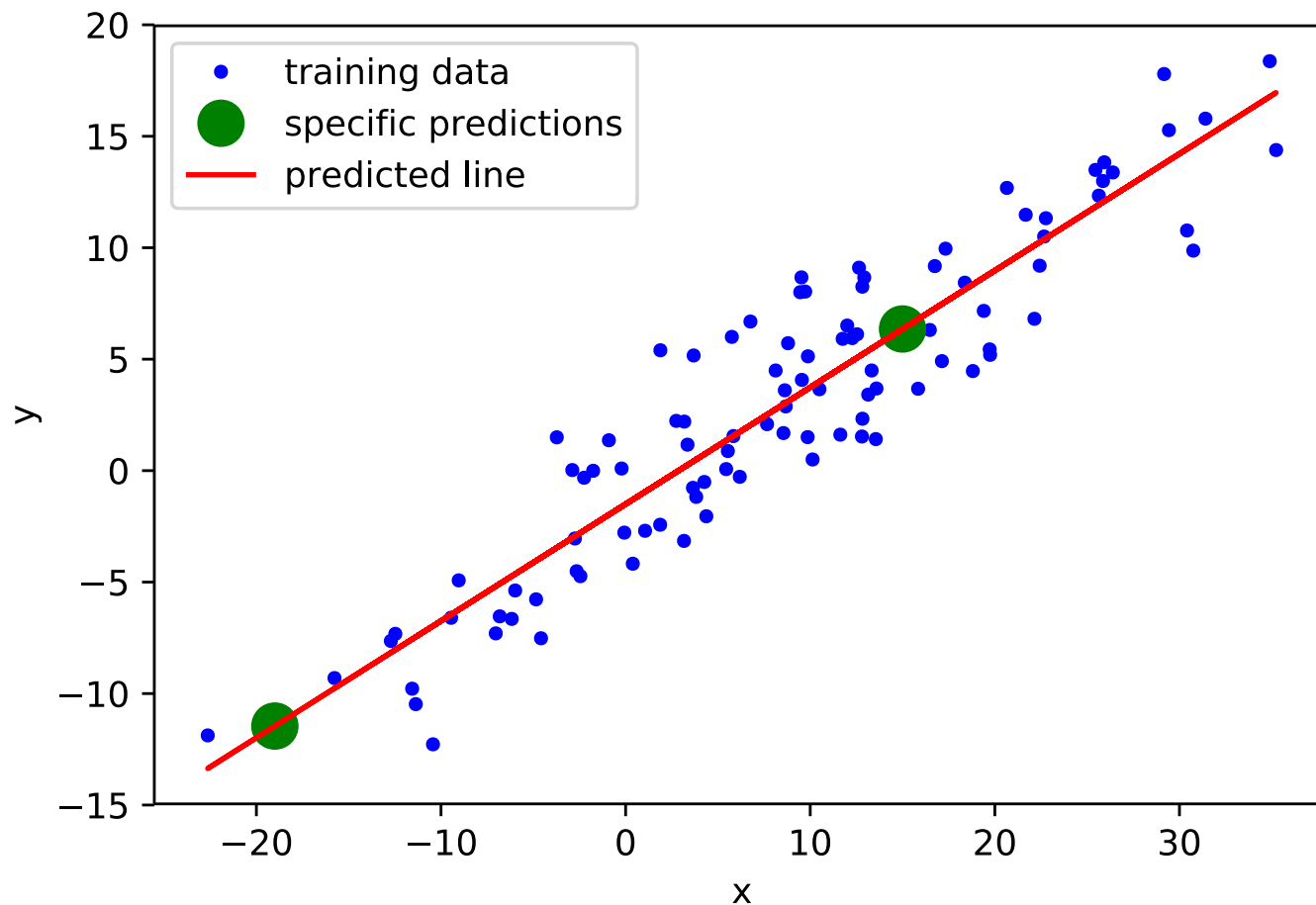


- This module focuses on Reactive AI (models, tools, techniques, interpretation, ...)
- Limited memory AI is gaining increasing adoption.
- Theory of Mind AI has been partially achieved (conversational AI can *mimic* a human's responses).
- Self-Aware AI is still science fiction.

Regression

Definition

Given data comprising a set of independent variables (of any type $\mathbf{x}$) with a set of dependent variables (numeric only $\mathbf{y}$), find the relationship $\mathbf{y} = f(\mathbf{x})$ having the maximum likelihood given the available observations $\{\mathbf{x}_i, \mathbf{y}_i\}$.



Classification

Definition

Given data comprising a set of independent variables (of any type $\mathbf{x}$) with a set of dependent variables (categorical $\mathbf{y}$ (labels)), find the relationship $\mathbf{y} = f(\mathbf{x})$ having the maximum likelihood given the available observations $\{\mathbf{x}_i, \mathbf{y}_i\}$.

There are many ways of representing f : a classification tree is shown here.

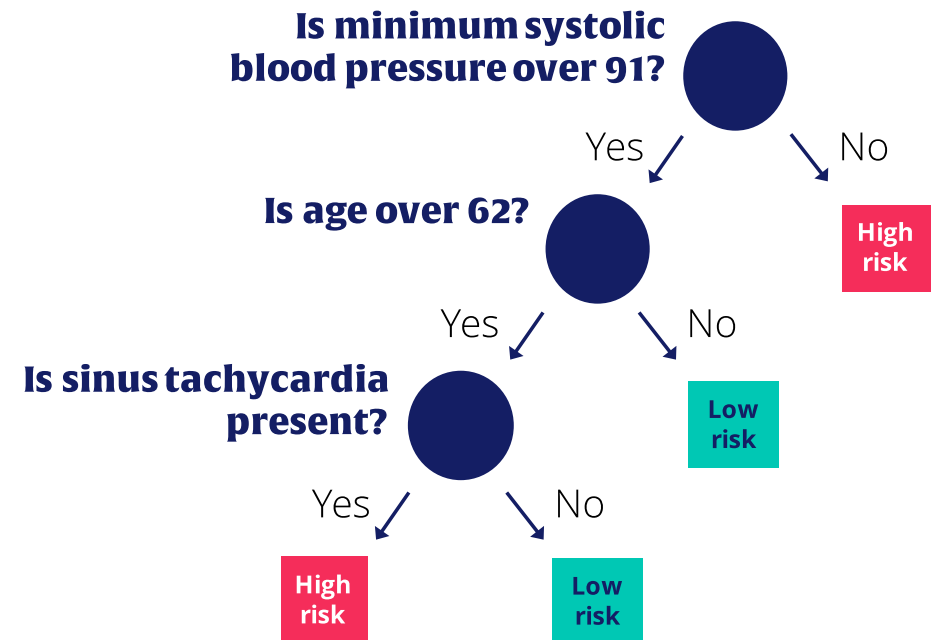


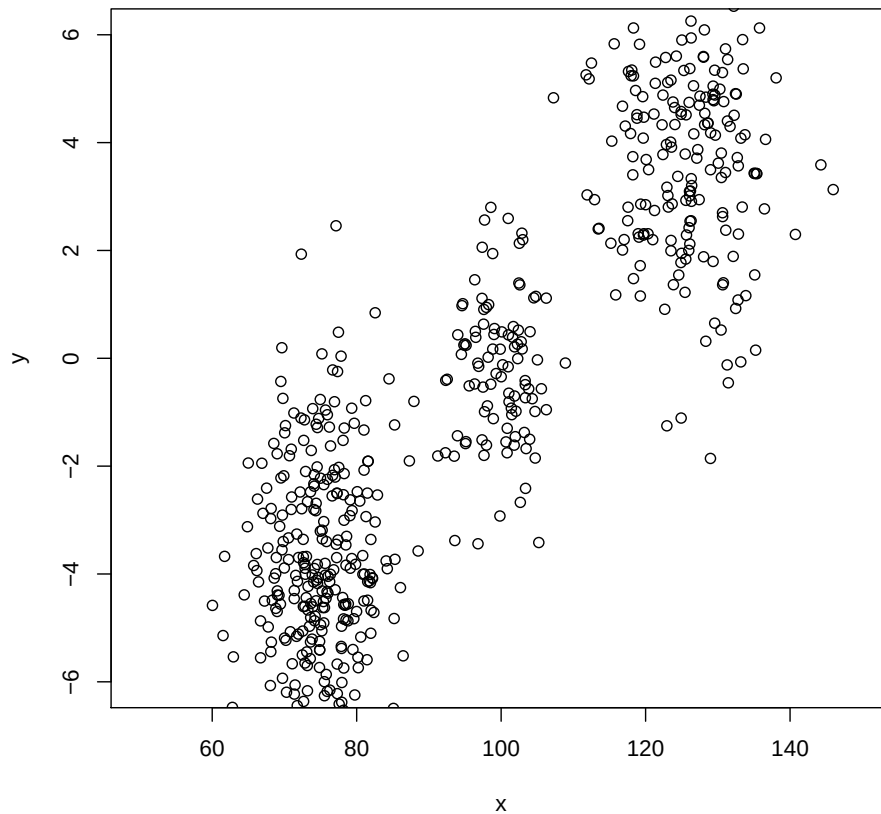
Diagram taken from the book Classification and Regression Trees, by Breiman L., Friedman J., Stone C. and Olshen R. 1984.

Clustering

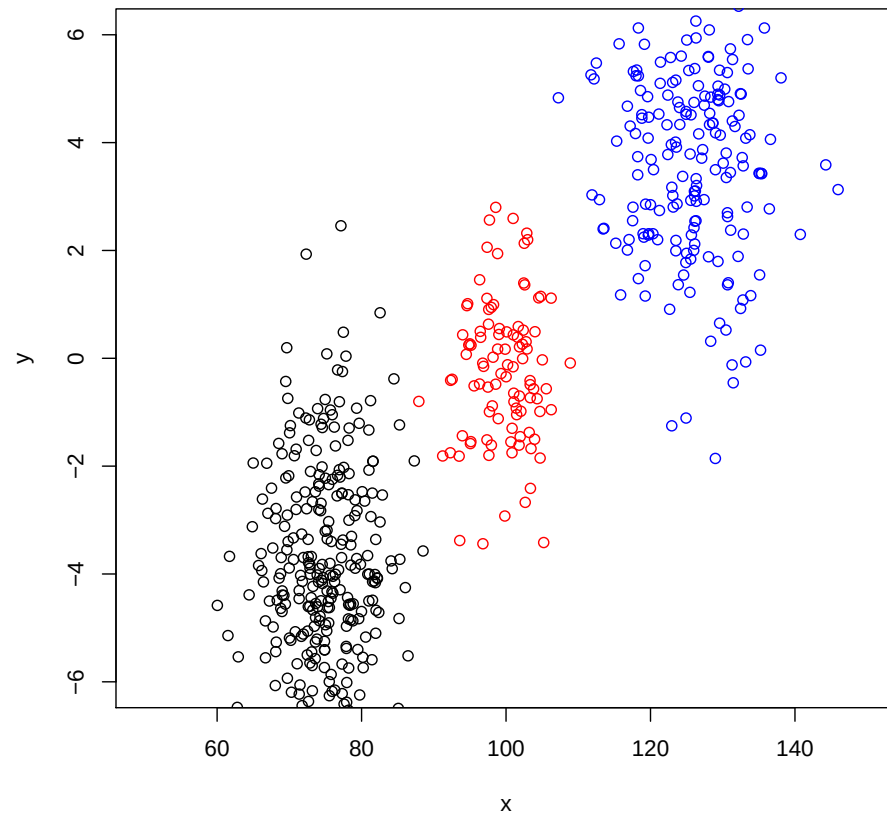
Definition

Clustering is the process of grouping data into classes or clusters, so that objects within a cluster have high similarity with each other but are dissimilar to objects in other clusters. Different similarity measures and/or algorithms result in different cluster arrangements.

Single cluster



Three clusters



Clustering Example 2: Image Analysis

Le et al (2012)

- Google Brain simulator crawled the web, looking for patterns in photographs on the web.
- It was *not* looking for anything in particular!
- One pattern came up strongly...

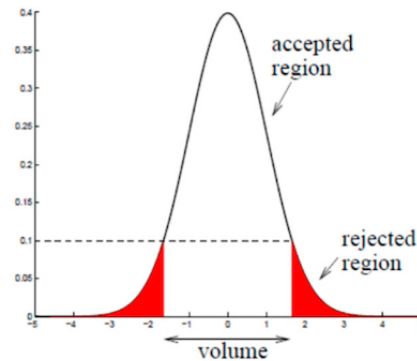
Supervised learning question: what does this represent?



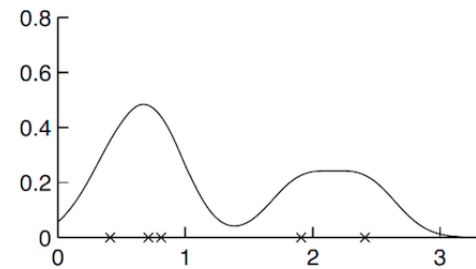
Anomaly Detection

Definition

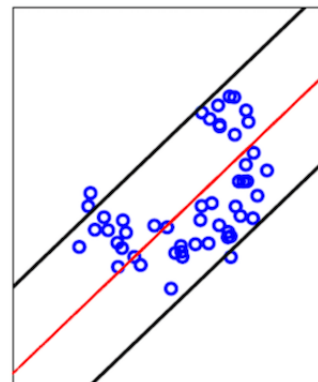
Anomaly detection identifies data points, events, and/or observations that depart from a dataset's normal behavior. Anomalous data can indicate problems, such as fraud, or opportunities, like a surge in demand for a product.



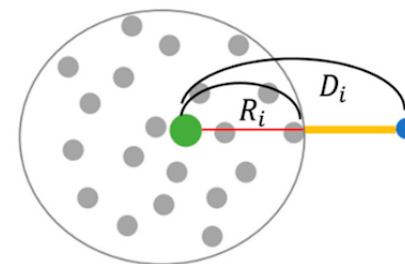
(a)



(b)



(c)



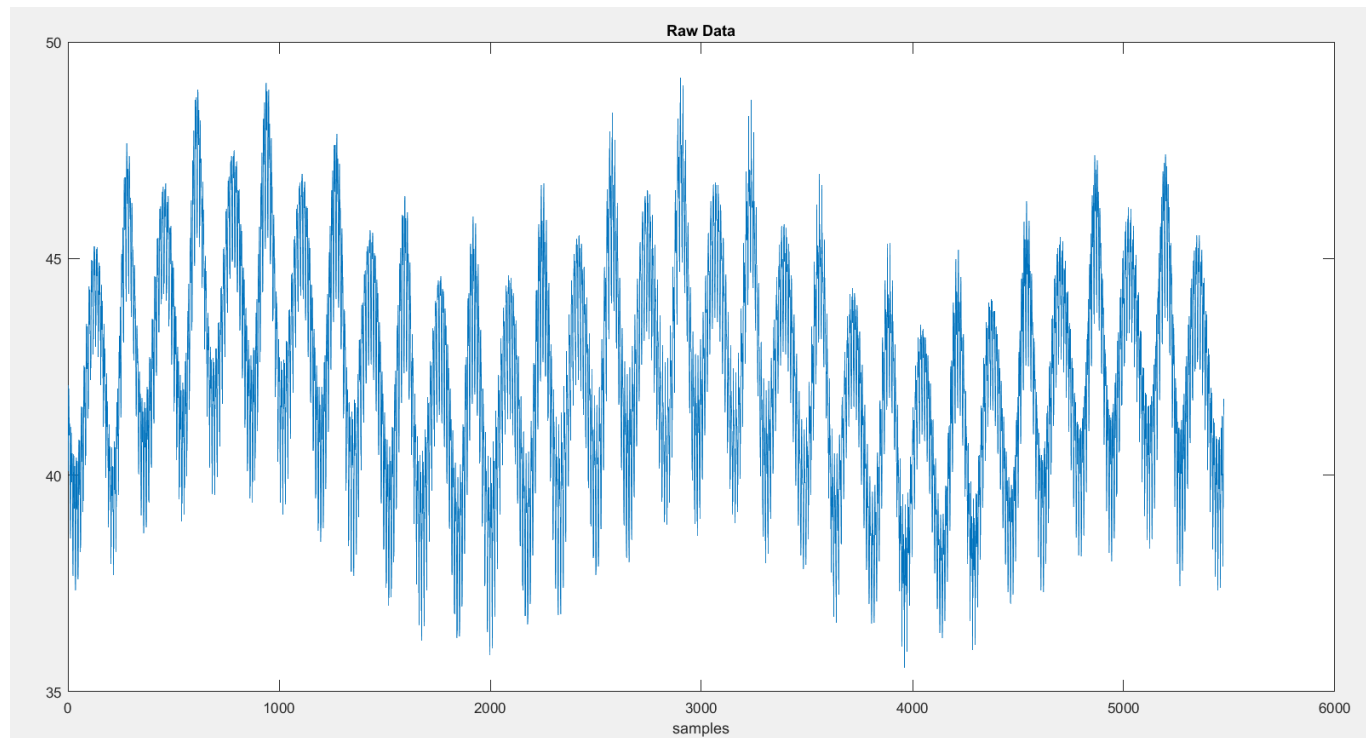
(d)

Source: Appl. Sci. 2019, 9, 4018; doi:10.3390/app9194018

Time Series Analysis

Definition

Time series data is a sequence of observations on the values that a variable taken at regularly-spaced time intervals. This data is sequentially correlated and techniques are needed to determine seasonality, trends, anomalies etc.



Source: <https://stats.stackexchange.com/q/458491>

Association Rules Mining

Definition

Frequent itemset mining looks for associations and correlations among items in large data sets. Associations are expressed as rules and quantified in terms of their *support* and *confidence*. The classical example is market basket analysis and the famous rule about buying diapers and beer together. See example transaction data below

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke



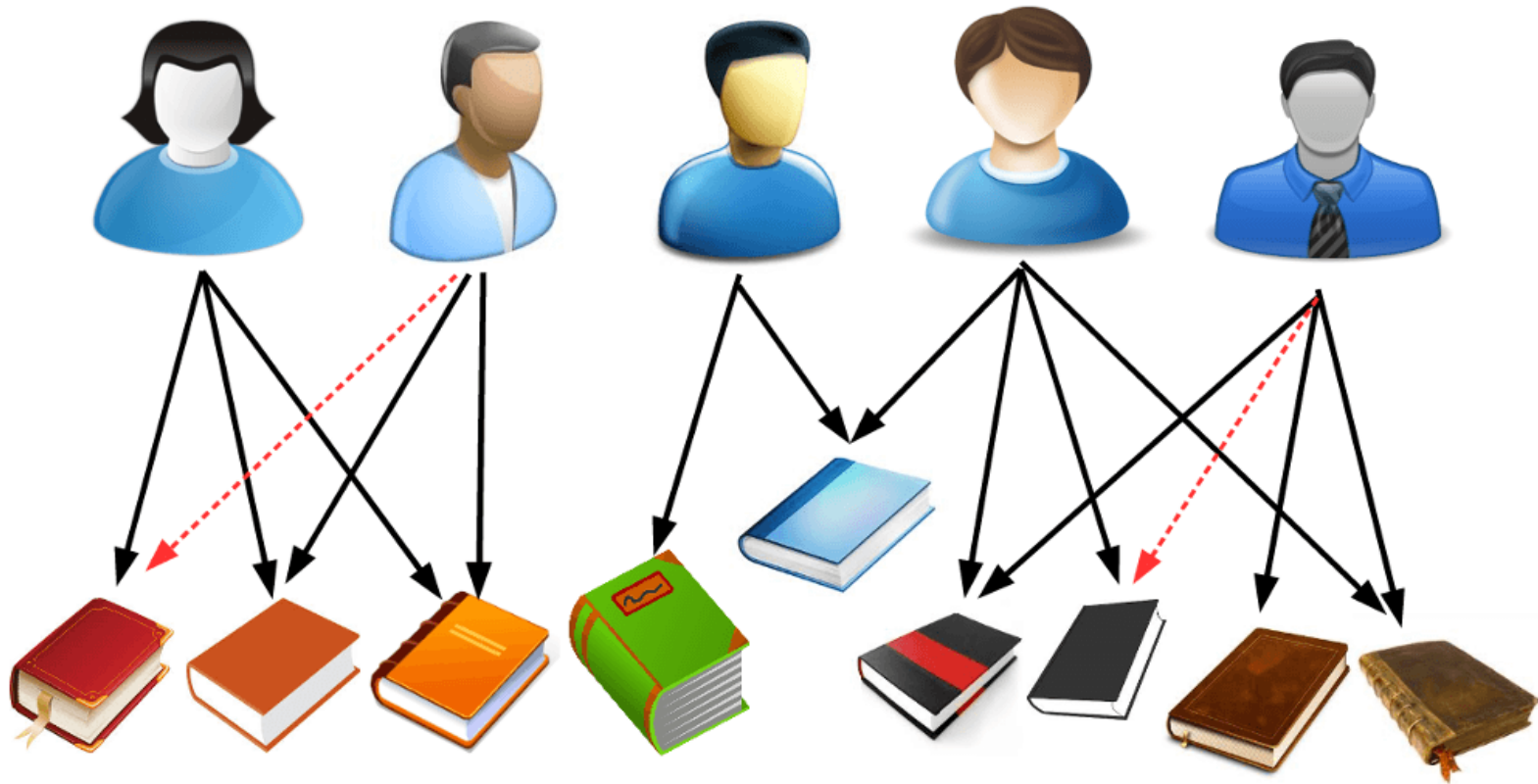
	Beer	Bread	Milk	Diaper	Eggs	Coke
T_1	0	1	1	0	0	0
T_2	1	1	0	1	1	0
T_3	1	0	1	1	0	1
T_4	1	1	1	1	0	0
T_5	0	1	1	1	0	1

ARM is a special case of structure mining, which includes graph mining, molecular and DNA nucleotide analysis

Recommender Systems

Definition

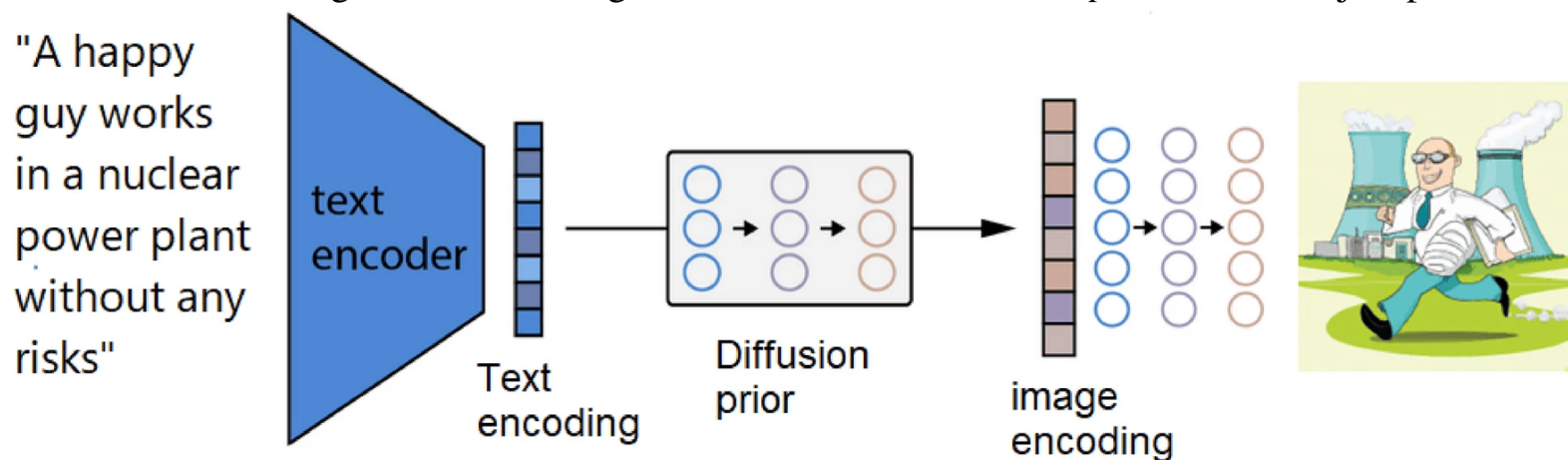
Collaborative filtering is an extension of item-based association rules mining to consider relationships between users and items. Generally, if two users have similar behaviour and/or preferences, items favoured by one user will also be favoured by the other. This can be used to recommend “new” items to users. Used very commonly on ecommerce websites for cross-selling.



Generating text and images

Definition

Text, images and other rich media can be encoded as numbers and used to train deep learning models that discover hidden relationships and can be used to predict next tokens in a sequence of words, and with suitable mapping techniques such as stable diffusion, can use stable diffusion to generate new images based on the learnt *latent space* instead of just pixels.



Traditional computing vs. Machine learning

➤ **Models** transform **input(s)** to **outputs**. **Example:** *Compound Interest*

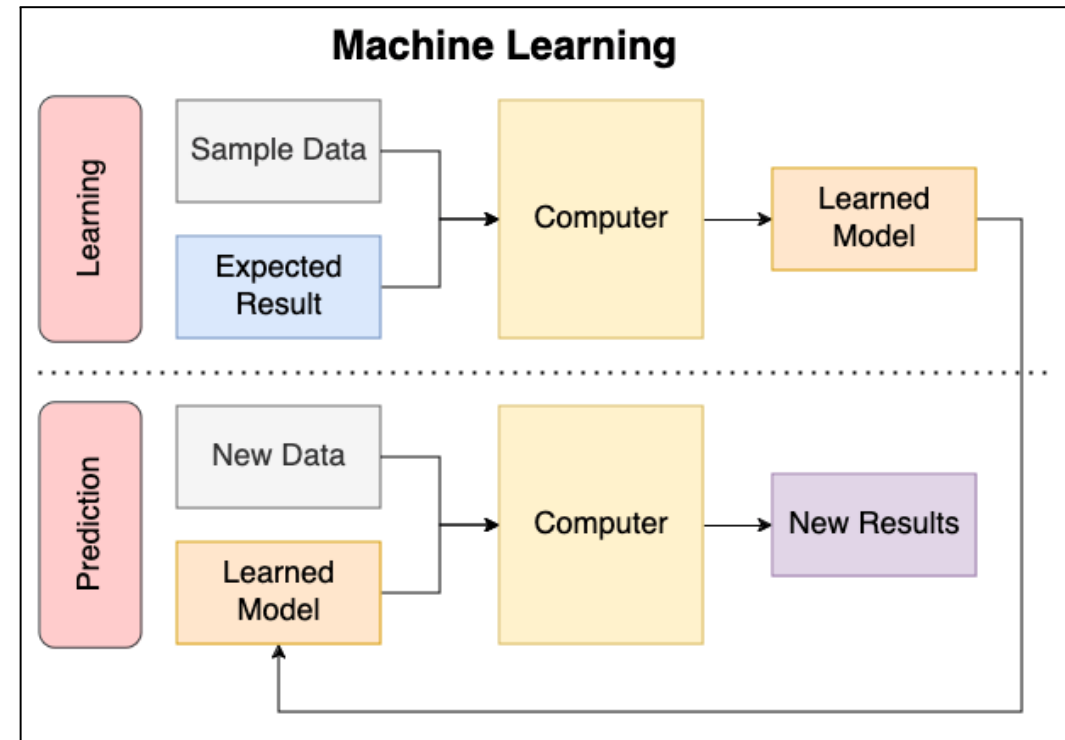
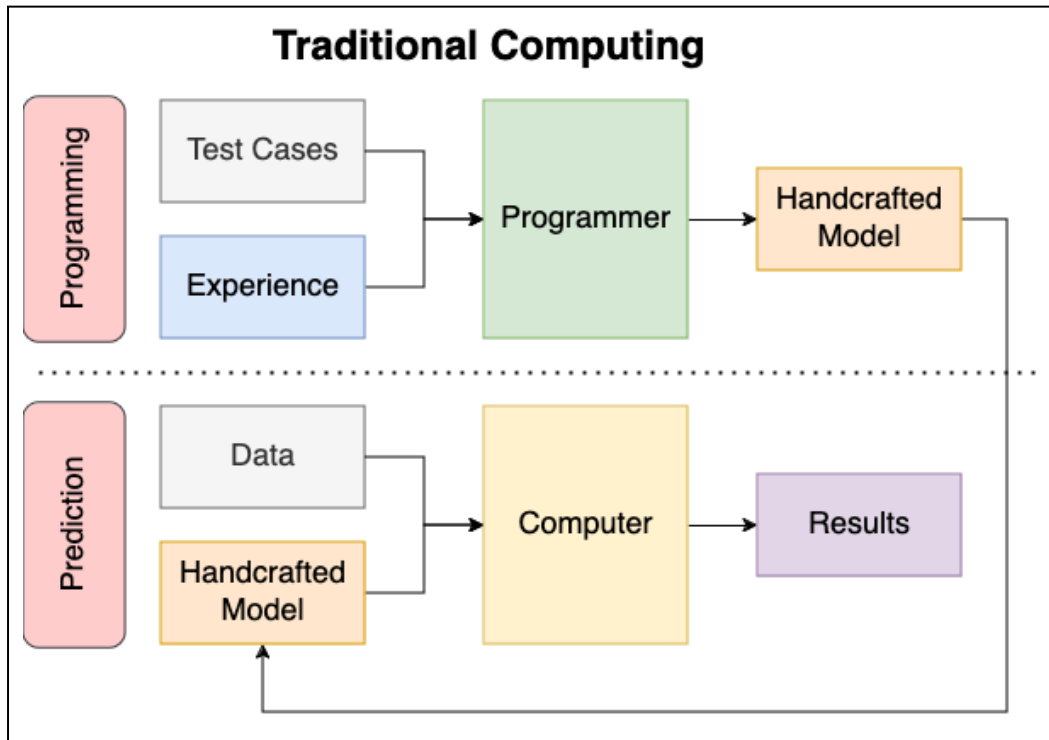
Traditional Computing: Implement a known model

- Given input values $P = \text{Principal (amount invested)}$, $r = \text{interest rate (as percentage)}$ and $n = \text{term (number of years invested)}$
- Output value is $I = P \left(1 + \frac{r}{100}\right)^n - P$ is the *compound interest*.
- Note that the model is provided *explicitly* and is programmed as such.
- To validate the implementation of the model: unit testing on specific examples.

Machine learning: Derive a model from data

- Given inputs P , r and n and associated compound interest output I , interpreted as before, but the formula for I is not known
- ... with several examples of inputs and associated output (training data)
- We *search* for a model that relates the given inputs to the outputs.
- To validate: use the model to predict the output from previously unseen (test) data.

Comparing the two paradigms



Some key cross-cutting concerns

- Key role of Exploratory Data analysis (EDA):
 - How can we standardise and clean up the data so it is ready?
 - How can we check assumptions and discover obvious/useful patterns?
 - How can we use EDA to start model formulation?
- Model Building as a process
 - Models are built from expressions evaluated on data - but how can we choose them?
 - How can we improve models?
 - How can we assess the quality of a model and report on/visualise this?
- Machine learning needs to be objective, repeatable, scalable and ethical

Sample rows from auto-mpg dataset

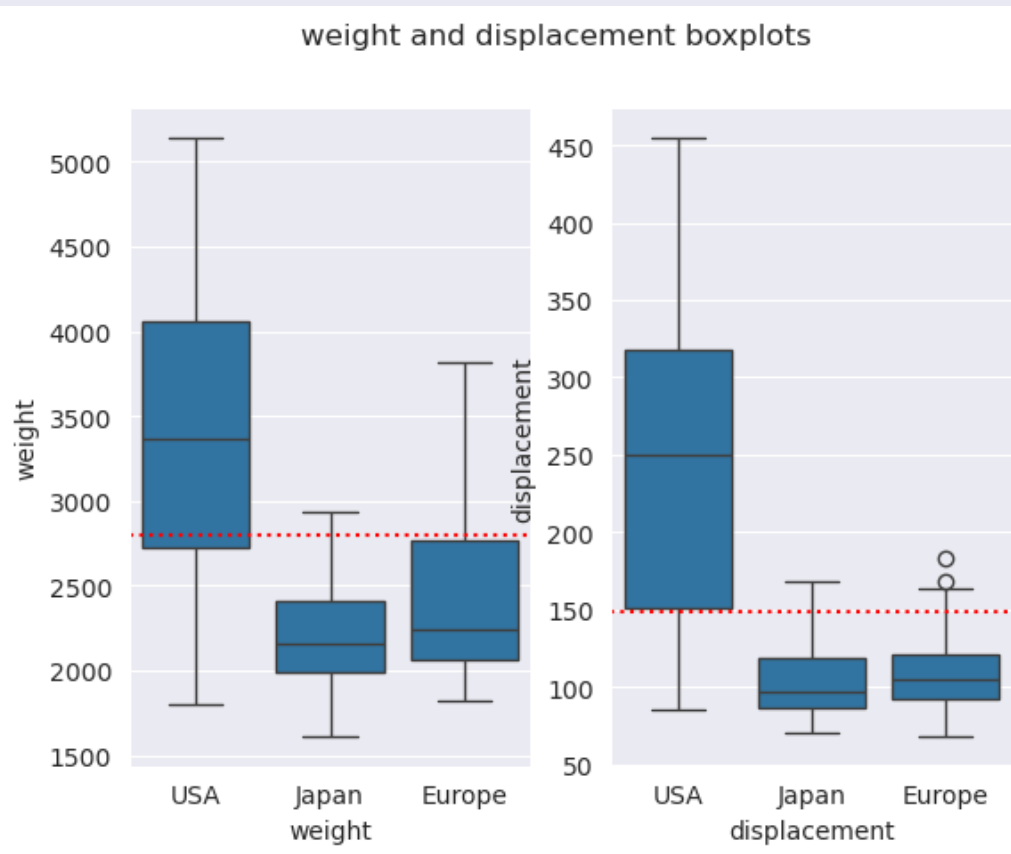
Pandas dataframe: sample rows used for training data

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
32	25.0	4	98.0	?	2046	19.0	71	1	ford pinto
112	19.0	4	122.0	85	2310	18.5	73	1	ford pinto
130	26.0	4	122.0	80	2451	16.5	74	1	ford pinto
168	23.0	4	140.0	83	2639	17.0	75	1	ford pinto
174	18.0	6	171.0	97	2984	14.5	75	1	ford pinto
206	26.5	4	140.0	72	2565	13.6	76	1	ford pinto

- The first (unnamed) “column” is the *dataframe index*; the values are like rowids in a database table.
- The rows contain data about automobiles in the US market in the early 1970s.
- The data is used to train models to predict “mpg” (*target*) given the other columns (features).

Example plot to develop an understanding of the data

Boxplot showing how manufacturer country affects weight and engine size



- Boxplot is one of many plot types available in plotting libraries
- Generated by a few lines of python code
- Side-by-side placement can help comparison
- What do the plots tell us?

What do we have to go on?

- We have data collected from a selection of cars, but our machine learning has no purpose yet.
- What if we wished to use this data to predict fuel efficiency (in mpg, our *target*) from the other fields (*features*) for a car that was not in this data set?
- What type of machine learning objective is that?
- What features are needed, what would the expressions look like, how would we choose the best model, how would we measure prediction performance?
- Such questions will be addressed in this module.

Resources



- URL: [Moodle module - the VLE for this module](#)
- Used for all notices, assignment briefs and practical work submissions.



- URL: [AI Engineering 1 \(2026 Semester 1\) pages on github.io](#)
- Used for content delivery (lecture notes and labs).



- URL: [Slack workspace for comments and Q&A relating to the module, TBA.](#)

Software

All software used during this module is open source or freely available for non-commercial use (full details given in notes). Primarily

- Anaconda (**Python 3.14, 64 bit**)
- scikit-learn
- pandas

www.anaconda.com

scikit-learn.org

pandas.pydata.org

Further Reading

Please note that the notes and labs we provide should be sufficient to pass to pass this module, so the books below are intended as *further reading*, not *recommended reading*.

Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python

by *Sebastian Raschka*, combines clear explanations of machine learning concepts with worked examples using traditional ML (Scikit-learn) and deep learning (PyTorch) frameworks.

An Introduction to Statistical Learning, with Applications in Python

by *G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor*, a classic text, written by authors who developed much of the theory, with well-chosen examples that use Python. Free to download.

Python for Everybody: Exploring Data using Python 3

by *Charles R Severance* (Available as PDF)

Useful summary of basic python concepts and a good introduction to the type of data manipulation can can be performed using core python data structures and idioms.

Python Data Science Handbook

by *Jake vanderPlas* (Available from website), is both a textbook and a set of freely-available Jupyter notebooks that go into more detail on implementing some of the material in this module using pandas etc.